Workspace ONE to Intune — macOS Device Migration

Implementation Solution Guide

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Executive Summary — Workspace ONE to Intune: macOS Device Migration

Moving a Mac fleet off Workspace ONE and onto Microsoft Intune is not a switch you flip — it is a device handoff. Each Mac must be re-assigned in Apple Business Manager from the old MDM server to Intune, re-enrolled, and then rebuilt to the same posture it had before: settings-catalog policies, custom `.mobileconfig` profiles, and shell scripts, all at parity. Done ad hoc, it produces half-managed Macs, dropped profiles, and unhappy users. Done the repeatable way, it is a controlled, ringed cutover you can prove on a pilot before you scale.

This is a complete, deploy-it-today package for migrating macOS devices from Workspace ONE to Microsoft Intune, written for an administrator with no prior migration experience. It treats Workspace ONE internals as the vendor's domain (deferred to VMware documentation) and gives exact, named steps for the Microsoft Intune and Apple Business Manager surfaces you control.

What you get

- A plain-language Solution Overview of the macOS migration model — Apple Business Manager re-assignment, ADE / Account-driven re-enrollment, and rebuilding configuration for parity.
- A Requirements Guide with verify / implement-if-missing for every prerequisite.
- An Implementation Guide of precise, ordered steps: connect ABM to Intune, build enrollment profiles, reassign Macs in ABM, re-enroll by ring, rebuild profile and shell parity, validate, expand.
- Five PowerShell scripts for migration-readiness discovery, per-device enrollment and parity validation, a migration status and parity report, safe remediation of stuck enrollments, and a migration-ring group builder.
- Testing and Troubleshooting guides with explicit pass/fail gates.

What this delivers

A repeatable Workspace ONE to Intune macOS migration: each Mac re-assigned in Apple Business Manager, re-enrolled into Intune via Automated Device Enrollment or Account-driven enrollment, and rebuilt to full configuration parity, rolled out in migration rings with a pilot gate and a documented rollback.

The standard deployment model

Assess to Configure to Pilot to Validate to Expand.

Outcome

Macs moved from Workspace ONE to Intune — re-assigned in Apple Business Manager, re-enrolled, and rebuilt to full macOS profile and shell-script parity — proven on a pilot ring before you scale.

1. Solution Overview — Workspace ONE to Intune: macOS Device Migration

What this migration is

Migrating macOS devices from VMware Workspace ONE UEM to Microsoft Intune means changing which MDM authority manages each Mac. For Apple devices that authority is decided in Apple Business Manager (ABM): an enrolled Mac is tied, through ABM, to one MDM server. Moving to Intune therefore requires three things to happen for each device — re-assign the Mac in ABM from the Workspace ONE MDM server to the Intune MDM server, re-enroll the Mac into Intune, and rebuild its configuration in Intune so the device ends up with the same posture it had under Workspace ONE.

Unlike a Windows migration, a Mac cannot simply switch authority in place. Apple's model ties the management profile to the device record in ABM, so re-enrollment is the mechanism — either Automated Device Enrollment (ADE, formerly DEP) for corporate-owned Macs, or Account-driven enrollment for user-initiated scenarios. The configuration is then rebuilt across the three macOS Intune surfaces: the settings catalog, custom `.mobileconfig` profiles, and shell scripts.

Why this is done this way (the rationale)

• A Mac obeys exactly one MDM server at a time. ABM is the system of record that decides which one, so re-assignment in ABM is the unavoidable hinge of any Mac MDM migration.

• Apple does not provide an in-place authority swap for macOS; re-enrollment is how the device picks up the new management profile. ADE makes that re-enrollment hands-off for corporate Macs.

• Configuration is not portable between MDMs. Workspace ONE profiles, baselines, and scripts do not transfer to Intune; the posture must be rebuilt natively as settings-catalog policies, custom `.mobileconfig` profiles, and shell scripts so it is supportable in the target.

• Macs are live, in-use production devices. A ringed cutover — pilot first, then expand — catches a parity gap or a re-enrollment failure on a few devices before it reaches the fleet.

How it works (end to end)

1. Connect ABM to Intune: add Intune as an MDM server in ABM and stand up the Automated Device Enrollment connection (ADE token) and the Apple MDM Push (APNs) certificate in Intune.

1. Build the macOS enrollment profile in Intune so re-enrolled Macs receive the right setup, management mode, and user experience.

1. Establish the parity target: inventory the current Workspace ONE macOS posture (profiles, restrictions, scripts) and map each item to its Intune surface — settings catalog, custom `.mobileconfig`, or shell script.

1. Build the parity objects in Intune (settings-catalog policies, custom profiles, shell scripts)

assigned to the migration groups, so configuration is waiting when a Mac arrives.

1. Reassign Macs in ABM, ring by ring, from the Workspace ONE MDM server to the Intune MDM server.

2. Re-enroll the Macs into Intune (ADE for corporate Macs; Account-driven for user-initiated), removing Workspace ONE management as part of the cutover.

1. Validate parity on the pilot ring — enrollment succeeded, profiles and scripts applied, posture matches — then expand ring by ring to the rest of the fleet.

The three macOS configuration surfaces (parity targets)

- Settings catalog — most modern macOS controls (Software Update, restrictions, login window, security and privacy) rebuilt as settings-catalog policies.
- Custom `.mobileconfig` profiles — any control with no settings-catalog payload, delivered as an uploaded Apple configuration profile.
- Shell scripts — controls enforceable only by a command, delivered as Intune macOS shell scripts on a re-enforcement schedule.

What you control in the target (Intune + ABM)

- The ABM-to-Intune connection (MDM server assignment, ADE token, APNs certificate).
- The macOS enrollment profile (setup steps, management mode, user experience).
- The rebuilt configuration parity (settings catalog, custom profiles, shell scripts).
- The migration rings (which Macs move in which wave) and the pilot gate.

What stays with the vendor (deferred)

- Workspace ONE UEM internals — how to release, unenroll, or de-assign a Mac on the Workspace ONE side, and the exact Workspace ONE console paths — are deferred to VMware Workspace ONE documentation. This solution drives the Intune and Apple Business Manager surfaces you own.

Limitations (be honest with buyers)

- Configuration does not migrate; it is rebuilt. Plan the parity mapping work explicitly.
- Re-enrollment is user-visible on Macs not enrolled via ADE; communications matter.
- ABM re-assignment and the Workspace ONE-side release are coordinated, not instantaneous — expect a brief coexistence window per device.
- FileVault recovery keys and certificate-based items may need re-issue under the new authority;

treat them as explicit parity items, not assumptions.

The standard deployment model

> *Connect ABM to Intune, build the enrollment profile, rebuild parity objects, then per ring: reassign in ABM, re-enroll, validate parity, expand. Model: Assess to Configure to Pilot to Validate to Expand, with a pilot-ring gate and a documented rollback.*

2. Requirements Guide — Workspace ONE to Intune: macOS Device Migration

Verify every requirement before configuring. Each entry: Description, Why, Verify, and Implement-if-missing. Category-wide prerequisites are in `global-requirements-migrations.md` (target Entra ID tenant, target Intune licensing/admin, source platform access, source inventory, enrollment prerequisites, pilot/wave structure, and change control/coexistence/rollback). The items below are the macOS-specific prerequisites for this migration.

2.1 Apple Business Manager access and the Intune MDM server

Description: An Apple Business Manager organization where you can manage MDM servers and device assignments, with Intune added as an MDM server. Why: ABM is the system of record that decides which MDM manages each Mac; you reassign devices here. Verify: Sign in to `https://business.apple.com` as an ABM Administrator or Device Enrollment Manager; under MDM servers, confirm an Intune MDM server entry exists. Implement: In ABM add a new MDM server and link it to Intune (see Implementation Step 1); ensure you hold a role that can reassign devices.

2.2 Apple MDM Push (APNs) certificate in Intune

Description: A valid, unexpired Apple MDM Push certificate connecting Intune to Apple's push service. Why: No Apple device — including macOS — can be managed by Intune without an active APNs certificate. Verify: Intune to Devices to Enrollment to Apple to Apple MDM Push certificate shows present and not near expiry. Run `WS1MAC-Discovery.ps1`. Implement: Devices to Enrollment to Apple to Apple MDM Push certificate; create/renew with the same Apple ID you will keep long term (never let it lapse — management stops if it does).

2.3 Automated Device Enrollment (ADE) token / enrollment program token

Description: An ADE (DEP) token in Intune that syncs the corporate Macs assigned to the Intune MDM server in ABM. Why: ADE is how corporate-owned Macs re-enroll hands-off after you reassign them in ABM. Verify: Intune to Devices to Enrollment to Apple to Enrollment program tokens shows a token with synced macOS devices. Run `WS1MAC-Discovery.ps1`. Implement: In ABM download the server token for the Intune MDM server; in Intune to Devices to Enrollment to Apple to Enrollment program tokens to Add, upload it and sync.

2.4 Source access: Workspace ONE UEM admin

Description: Administrative access to the Workspace ONE UEM console for the in-scope Macs. Why: You must inventory the current macOS posture and release/unenroll devices on the source side during cutover. Verify: You can sign in to the Workspace ONE UEM console and export the macOS device list, profiles, and scripts for the migration scope. Implement: Obtain the Workspace ONE admin role from the platform owner. (Workspace ONE console paths and unenroll mechanics are deferred to VMware Workspace ONE documentation.)

2.5 The macOS parity target (current posture inventory)

Description: A documented inventory of every Workspace ONE macOS profile, restriction, payload, and script that must be reproduced in Intune, each mapped to its Intune surface. Why: Parity is the goal of the migration; you cannot rebuild what you have not inventoried and mapped. Verify: You have a mapping table — each Workspace ONE item to settings catalog, custom `.mobileconfig`, or shell script. Implement: Export the macOS configuration from Workspace ONE; sort each item into one of the three Intune surfaces (see Implementation Step 4). Record the parity tag `WS1MAC-Parity-vX`.

2.6 Custom `.mobileconfig` source files (where needed)

Description: The Apple configuration profile files for any control with no settings-catalog payload. Why: Some payloads exist only as custom profiles; without the file you cannot reach parity. Verify: For each control mapped to "custom profile", you have the `.mobileconfig` file ready to upload. Implement: Obtain or author the `.mobileconfig` (Apple Configurator, a profile tool, or export from the source) for each such control.

2.7 Migration ring groups

Description: Microsoft Entra security groups defining the migration rings — at minimum a pilot ring and one or more expansion rings. Why: The cutover is ringed; the pilot ring is the gate before fleet migration. Verify: Entra to Groups shows the migration ring groups. Run `WS1MAC-Set-MigrationRingTargeting.ps1` to build/populate them. Implement: Entra to Groups to New group to Security; create the rings named per Step 3, or run `WS1MAC-Set-MigrationRingTargeting.ps1`.

2.8 User communications plan

Description: A user-facing notice covering what will happen on the Mac, when, and any user action (for non-ADE re-enrollment) plus support contact. Why: macOS re-enrollment can be user-visible; clear comms cut tickets and failed enrollments. Verify: A comms message and schedule exist per ring. Implement: Draft a short per-ring notice (timing, what to expect, who to contact) before cutover.

Checklist

- Apple Business Manager access + Intune added as an MDM server
- Apple MDM Push (APNs) certificate active — `WS1MAC-Discovery.ps1`
- ADE / enrollment program token synced — `WS1MAC-Discovery.ps1`
- Workspace ONE UEM admin access (source)

- macOS parity target inventoried and mapped to Intune surfaces
- Custom `.mobileconfig` source files ready where needed
- Migration ring groups created — `WS1MAC-Set-MigrationRingTargeting.ps1`
- User communications plan per ring

3. Implementation Guide — Workspace ONE to Intune: macOS Device Migration

Follow these steps in order. Intune steps are in the Microsoft Intune admin center (`https://intune.microsoft.com`); Apple Business Manager steps are at `https://business.apple.com`. Workspace ONE console steps (releasing/unenrolling a Mac on the source side) are deferred to VMware Workspace ONE documentation and flagged where they occur.

> *Model: Assess to Configure to Pilot to Validate to Expand. Connect ABM to Intune and rebuild the macOS posture once; then per ring reassign in ABM, re-enroll, validate parity, and expand. The pilot ring is a gate — do not expand until it passes (Step 9) and keep the rollback (Step 8.6) ready.*
>
> *Naming: every artifact is named* `<Purpose> - <Scope> - <Stage>` *and reused verbatim across the docs and scripts. Migration rings:* `WS1MAC Migration - Pilot Ring` *and* `WS1MAC Migration - Production Ring`*. Parity objects carry the parity tag* `WS1MAC-Parity-vX` *in their description.*

Step 1 — Connect Apple Business Manager to Intune (Assess to Configure)

1. Confirm the Apple MDM Push (APNs) certificate: Intune to Devices to Enrollment to Apple to

Apple MDM Push certificate; present and not near expiry (Requirement 2.2). Run `WS1MAC-Discovery.ps1`.

1. In Apple Business Manager to your account to Preferences (or Settings) to MDM servers, add a new

MDM server named `WS1MAC Intune - macOS MDM Server`. Download its public key from Intune first if prompted: Intune to Devices to Enrollment to Apple to Enrollment program tokens to Add, which provides the public key to upload in ABM, then download the ABM server token.

1. In Intune to Devices to Enrollment to Apple to Enrollment program tokens to Add, upload the ABM

server token and complete the connection. Confirm the token appears and that synced macOS device count is non-zero once Macs are assigned (Requirement 2.3). Run `WS1MAC-Discovery.ps1`.

Step 2 — Build the macOS enrollment profile (Configure)

1. Intune to Devices to Enrollment to Apple to Enrollment program tokens to (open the token

`WS1MAC Intune - macOS MDM Server`) to Profiles to Create profile to macOS.

1. Name: `WS1MAC Enrollment - macOS - ADE`. Description: include the parity tag `WS1MAC-Parity-vX`.

Next.

1. Configure the enrollment experience: management mode (supervised for corporate Macs), Setup Assistant screens to skip/show, locked enrollment per policy, and the user experience. Next.

1. Review and Create. This profile is what re-enrolled corporate Macs receive after ABM reassignment.

2. (User-initiated Macs) For Macs that will use Account-driven enrollment instead of ADE, confirm the Account-driven enrollment path is enabled per your tenant; name that flow `WS1MAC Enrollment - macOS - AccountDriven` in your runbook. Account-driven enrollment specifics are configured per the current Intune Apple enrollment options.

Step 3 — Create the migration ring groups (Configure)

1. Entra to Groups to New group to Security to name `WS1MAC Migration - Pilot Ring` to add a few representative Macs. Create.

1. Create `WS1MAC Migration - Production Ring` for the rest of the in-scope fleet. Create.

2. Or run `WS1MAC-Set-MigrationRingTargeting.ps1` to create and populate these groups from a device list.

Step 4 — Establish the parity target and map each item (Assess to Configure)

1. From Workspace ONE, export the current macOS posture: profiles, restrictions, payloads, and scripts for the in-scope Macs (Requirement 2.5). (Workspace ONE export mechanics are deferred to VMware Workspace ONE documentation.)

1. Sort each item into one of the three Intune surfaces:

• settings catalog — most controls (Software Update, restrictions, login window, security and privacy), built in Step 5.

• custom `.mobileconfig` — controls with no settings-catalog payload, built in Step 6.

• shell script — controls enforceable only by a command, built in Step 7.

1. Record the mapping table and the parity tag `WS1MAC-Parity-vX`; this tag goes in every parity object's description so the set is traceable.

Step 5 — Build the settings-catalog parity policy (Configure)

1. Intune to Devices to macOS to Configuration to Create to New Policy.

2. Platform: macOS. Profile type: Settings catalog. Create.

3. Name: `WS1MAC Parity - macOS Settings - Pilot`. Description: parity tag `WS1MAC-Parity-vX`. Next.

4. Click Add settings. For each mapped control from Step 4, search the settings picker, add it, and

set the value to match the Workspace ONE posture (the source values are the parity reference).

1. Next to Assignments to Add groups to `WS1MAC Migration - Pilot Ring`. Next to Create.

Step 6 — Build the custom `.mobileconfig` parity profiles (Configure, only if needed)

1. For each control with no settings-catalog payload (from Step 4), have the `.mobileconfig` file

ready (Requirement 2.6).

1. Intune to Devices to macOS to Configuration to Create to New Policy to Platform: macOS to Profile

type: Templates to Custom to Create.

1. Name: `WS1MAC Parity - macOS Custom - Pilot`. Description: parity tag. Provide a deployment channel

and upload the `.mobileconfig` configuration profile file.

1. Assignments to Add groups to `WS1MAC Migration - Pilot Ring`. Create.

Step 7 — Build the shell-script parity (Configure, only for script-only controls)

1. For each control enforceable only by a command (from Step 4): Intune to Devices to macOS to Shell

scripts to Add.

1. Name: `WS1MAC Parity - macOS Shell - Pilot`. Description: parity tag. Upload the shell script that

implements the control.

1. Script settings: run as root = Yes where required; set a run frequency so the control re-enforces

on a schedule (re-applying corrects drift). Next.

1. Assignments to Add groups to `WS1MAC Migration - Pilot Ring`. Add to Next to Add.

Step 8 — Migrate the pilot ring: reassign in ABM, re-enroll, with rollback ready (Pilot)

1. Confirm the parity objects (Steps 5 to 7) are assigned to `WS1MAC Migration - Pilot Ring` and the

enrollment profile `WS1MAC Enrollment - macOS - ADE` is assigned to the token. Run `WS1MAC-Validation.ps1` against a pilot Mac to confirm targeting.

1. In Apple Business Manager, select the pilot Macs and reassign their MDM server from the Workspace

ONE MDM server to `WS1MAC Intune - macOS MDM Server`. In Intune, sync the enrollment program token (Enrollment program tokens to the token to Sync) and confirm the pilot Macs now appear as assigned to Intune.

1. Release the pilot Macs from Workspace ONE management on the source side so they can re-enroll cleanly. (The exact Workspace ONE unenroll/release steps are deferred to VMware Workspace ONE documentation.)

1. Re-enroll the pilot Macs into Intune:

• Corporate (ADE) Macs: erase/reset or trigger re-enrollment so the Mac runs Setup Assistant and picks up `WS1MAC Enrollment - macOS - ADE` automatically.

• User-initiated Macs: have the user complete Account-driven enrollment per the comms (Step 2.5, Requirement 2.8).

1. Confirm enrollment: Intune to Devices to macOS to macOS devices shows each pilot Mac enrolled and managed by Intune. Run `WS1MAC-Validation.ps1` per device.

1. Rollback (if a pilot Mac fails to enroll or a parity gap blocks use): reassign that Mac's MDM server in ABM back to the Workspace ONE MDM server and re-enroll it to the source, or remove the Intune assignment of the offending parity object and sync. Record the failure and resolve before retrying. ABM reassignment back to the source is the fastest safe rollback for an individual Mac.

Step 9 — Validate parity on the pilot, then gate (Validate)

1. Run `WS1MAC-Validation.ps1` per pilot Mac: enrollment succeeded, settings-catalog policy, custom profiles, and shell scripts all applied.

1. Run `WS1MAC-Report.ps1` for the pilot ring: migration status and per-object parity (succeeded vs drift) across the pilot Macs.

1. Soak the pilot: have pilot users run their real apps and workflows for several business days; watch for parity gaps (missing restriction, login window, FileVault, certificate, app config) and re-enrollment issues. Complete the Testing Guide (document 5).

1. Gate: do not expand until the pilot passes the Testing Guide gate AND no unresolved enrollment or parity gap remains.

Step 10 — Expand to production (Expand)

1. Duplicate each piloted parity object (Intune to policy to Duplicate where available) and rename the copies `WS1MAC Parity - macOS Settings - Production`, `WS1MAC Parity - macOS Custom - Production`, and `WS1MAC Parity - macOS Shell - Production`, keeping the parity tag. (For shell scripts, re-create with the production name.) Assign them to `WS1MAC Migration - Production Ring`.

1. Migrate the production ring in waves: reassign each wave's Macs in ABM to `WS1MAC Intune - macOS MDM Server`, release them from Workspace ONE (deferred to VMware docs), sync the token, and re-enroll (Step 8.2 to 8.5).

1. Re-run `WS1MAC-Report.ps1` after each wave to confirm enrollment and parity and to catch drift; use `WS1MAC-Remediation.ps1` (default preview) for stuck enrollments.

1. Decommission: once a Mac is confirmed enrolled in Intune at parity, remove it from Workspace ONE per your change record.

Order of operations

```
Connect ABM to Intune to Enrollment profile to Ring groups to Map parity to
Settings-catalog parity to Custom .mobileconfig parity to Shell parity to
Pilot (reassign in ABM, re-enroll, rollback ready) to Validate parity + gate
to Expand in waves
```

5. Testing Guide — Workspace ONE to Intune: macOS Device Migration

Run after the pilot ring migrates (Implementation Step 8) and again per production wave (Step 10). Tests use the named artifacts from the Implementation Guide.

Test 1 — ABM reassignment took effect

Do: In Apple Business Manager, confirm the pilot Macs show MDM server `WS1MAC Intune - macOS MDM Server`; in Intune to Devices to Enrollment to Apple to Enrollment program tokens, sync the token and confirm the Macs are assigned to Intune. Expected: Each pilot Mac is assigned to the Intune MDM server in ABM and appears under the Intune enrollment token.

Test 2 — Re-enrolled into Intune

Do: Intune to Devices to macOS to macOS devices; or run `WS1MAC-Validation.ps1` per device. Expected: Each pilot Mac is enrolled and managed by Intune (no longer managed by Workspace ONE).

Test 3 — Settings-catalog parity applied

Do: Intune to Devices to macOS to Configuration to `WS1MAC Parity - macOS Settings - Pilot` to Device status / Per-setting status; or `WS1MAC-Validation.ps1`. Expected: Pilot Macs show Succeeded; the settings match the Workspace ONE parity reference.

Test 4 — Custom profiles and shell scripts applied

Do: Check `WS1MAC Parity - macOS Custom - Pilot` (Configuration to Device status) and `WS1MAC Parity - macOS Shell - Pilot` (Devices to macOS to Shell scripts to the script to Device status); or `WS1MAC-Validation.ps1`. Expected: Custom profiles show Succeeded and shell scripts show their last run as success.

Test 5 — Posture parity (real use)

Do: On a pilot Mac, confirm the migrated posture matches the prior Workspace ONE state — restrictions, login window, Software Update, FileVault, certificates, and any app config — by using the device for real work. Expected: No regression versus the Workspace ONE posture; users can do their normal work.

Test 6 — Fleet migration + parity report

Do: Run `WS1MAC-Report.ps1` for the ring. Expected: A clear per-device migration status and per-object parity breakdown (succeeded vs drift) for the ring.

Test 7 — Rollback rehearsed (pilot only)

Do: On one pilot Mac, rehearse the rollback (Implementation Step 8.6): reassign in ABM back to the Workspace ONE MDM server, or remove an Intune parity assignment and sync. Expected: The rollback path works and is documented before any production wave.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1 to 6 pass on the pilot ring; rollback (Test 7) is rehearsed; no unresolved enrollment or parity gap
Wave pass	Tests 1 to 4 and 6 pass for each production wave before the next wave migrates

6. Troubleshooting Guide — Workspace ONE to Intune: macOS Device Migration

Use `WS1MAC-Discovery.ps1` for tenant/enrollment readiness, `WS1MAC-Validation.ps1` for per-device enrollment and parity, `WS1MAC-Report.ps1` for fleet/ring status, and `WS1MAC-Remediation.ps1` for safe sync-based fixes (preview first). Workspace ONE-side issues (release/unenroll) are deferred to VMware Workspace ONE documentation.

A. Mac will not re-enroll into Intune

Symptom	Cause	Remediation
ADE Mac does not pick up the enrollment profile	Profile not assigned to the token, or token not synced after ABM reassignment	Confirm `WS1MAC Enrollment - macOS - ADE` is assigned to the token; sync the token; run `WS1MAC-Discovery.ps1`
Mac still managed by Workspace ONE	Source management not released before re-enrollment	Release the Mac from Workspace ONE (deferred to VMware docs), then re-enroll
Enrollment fails with push errors	APNs certificate expired or wrong Apple ID	Renew the APNs certificate with the long-term Apple ID; never let it lapse

B. ABM reassignment not reflecting in Intune

Symptom	Cause	Remediation
Reassigned Macs do not appear under the Intune token	Token not synced, or wrong MDM server selected in ABM	In ABM confirm the Mac's MDM server is `WS1MAC Intune - macOS MDM Server`; in Intune sync the enrollment program token
Devices show under the old token	They were assigned to the prior MDM server	Reassign in ABM, then sync; the device moves on the next sync

C. Configuration parity gaps after enrollment

Symptom	Cause	Remediation
A setting/restriction is missing	Control not mapped, or in the wrong Intune surface	Re-check the Step 4 mapping; add it to `WS1MAC Parity - macOS Settings - Pilot`, a custom profile, or a shell script as appropriate
Custom profile not applied	Wrong deployment channel or malformed `.mobileconfig`	Re-upload a valid `.mobileconfig` to `WS1MAC Parity - macOS Custom - Pilot`; confirm the channel
Shell-script control drifts back	No re-enforcement schedule	Set a run frequency on `WS1MAC Parity - macOS Shell - Pilot`; run `WS1MAC-Remediation.ps1` to re-sync

D. Object shows Error / not applied

- Confirm the Mac is enrolled and in the assigned ring group; run `WS1MAC-Validation.ps1`.
- Run `WS1MAC-Remediation.ps1` (preview, then `-WhatIf:$false`) to issue a device sync so the parity

objects re-evaluate.

- Resolve any conflicting policy that targets the same payload.

E. FileVault / certificate continuity

- FileVault recovery keys and certificate-based items do not carry over with the device; treat each as

an explicit parity item and re-issue under Intune. If FileVault escrow is missing post-migration, confirm the FileVault policy is assigned and the Mac has re-escrowed its key.

F. Coexistence confusion (two authorities)

• A Mac briefly managed by both during cutover should be released from Workspace ONE promptly. If a Mac is stuck dual-managed, release the source first, then re-sync Intune.

G. When to escalate

If a Mac is assigned to `WS1MAC Intune - macOS MDM Server` in ABM, released from Workspace ONE, the token is synced, and the device still will not enroll or apply parity after a sync, collect the device enrollment state from `WS1MAC-Validation.ps1` and open a Microsoft support case (or, for the source release behavior, a VMware Workspace ONE support case).

7. Best Practices — Workspace ONE to Intune: macOS Device Migration

• Migrate in rings, always. A small `WS1MAC Migration - Pilot Ring` proves ABM reassignment, re-enrollment, and parity before the `WS1MAC Migration - Production Ring` fleet moves — the single most important practice.

• Build parity before you reassign. Have the settings-catalog policy, custom `.mobileconfig` profiles, and shell scripts assigned to the ring so configuration is waiting the moment a Mac re-enrolls — not scrambled together after.

• Inventory and map the source posture first. Export the Workspace ONE macOS configuration and map every item to its Intune surface (settings catalog, custom profile, shell script) with the parity tag `WS1MAC-Parity-vX`; you cannot rebuild what you have not catalogued.

• Reassign in ABM and release from the source as one coordinated move. ABM reassignment plus the Workspace ONE-side release is what lets the Mac re-enroll cleanly; doing one without the other leaves a device half-managed.

• Prefer ADE for corporate Macs. Automated Device Enrollment makes re-enrollment hands-off and consistent; reserve Account-driven enrollment for genuinely user-initiated cases.

• Keep the APNs certificate and ADE token healthy. An expired APNs certificate stops all Apple management; a stale token will not sync reassigned Macs. Check both with `WS1MAC-Discovery.ps1` before each wave.

• Treat FileVault keys and certificates as explicit parity items. They do not carry over with the device; plan their re-issue under the new authority rather than assuming continuity.

• Communicate per ring. Tell users what re-enrollment looks like, when it happens, and who to contact; clear comms cut tickets and failed enrollments, especially for non-ADE Macs.

• Keep the rollback ready. The fastest safe rollback for a single Mac is to reassign it in ABM back to

the Workspace ONE MDM server; rehearse it on the pilot before any production wave.

- Validate parity, do not assume it. Run `WS1MAC-Validation.ps1` per device and `WS1MAC-Report.ps1`

per ring after every wave, and chase any drift with `WS1MAC-Remediation.ps1` (preview first).

- Decommission only after parity is confirmed. Remove a Mac from Workspace ONE only once it is enrolled in Intune at full parity, so you always have a coexistence fallback.

8. Deliverables Summary — Workspace ONE to Intune: macOS Device Migration

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The macOS migration model, why ABM reassignment + re-enrollment + parity rebuild, the three config surfaces, limitations
2. Requirements Guide	Every macOS-specific prerequisite with Verify / Implement-if-missing
Global Requirements (Migrations)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps: connect ABM to Intune, enrollment profile, ring groups, parity rebuild, reassign + re-enroll, validate, expand
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`WS1MAC-Discovery.ps1`	Migration-readiness: APNs certificate, ADE/enrollment token + synced Macs, enrolled macOS count/version spread, existing macOS config policies (read-only, Graph)
`WS1MAC-Validation.ps1`	Per-device enrollment + parity check: managed by Intune, settings-catalog/custom/shell objects applied (read-only, Graph)

Script	What it does
`WS1MAC-Report.ps1`	Fleet/ring migration status + per-object parity (succeeded vs drift) with optional CSV (read-only, Graph)
`WS1MAC-Remediation.ps1`	Safe remediation of stuck enrollments/drift — issues a device sync so parity objects re-apply (defaults to -WhatIf)
`WS1MAC-Set-MigrationRingTargeting.ps1`	Builds/populates the migration ring groups (`WS1MAC Migration - Pilot Ring` / `... - Production Ring`) from a device list (defaults to -WhatIf)

What the buyer receives

- This implementation guide as a branded Word document and PDF.
- All five PowerShell scripts in a single ZIP.
- A repeatable process following the model: Assess to Configure to Pilot to Validate to Expand.

Outcome

Macs moved from Workspace ONE to Intune — re-assigned in Apple Business Manager, re-enrolled, and rebuilt to full macOS profile and shell-script parity — proven on a pilot ring before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WS1MAC-Discovery.ps1

Purpose: Reports whether the Intune/Apple side is ready to receive Macs migrating off Workspace ONE: that an Apple MDM Push (APNs) certificate exists and is not near expiry, that an Automated Device Enrollment (ADE) / enrollment program token is present with macOS devices synced, how many Macs are already enrolled and their macOS version spread, and which existing macOS configuration policies exist (so you can plan parity and spot conflicts). Read-only against Microsoft Graph (Intune). Workspace ONE/ABM-side state is checked in those consoles and is deferred to vendor documentation. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WS1MAC-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and these read scopes (consented at sign-in): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS APNs certificate presence/expiry, ADE/enrollment token(s) with synced macOS counts, count and version spread of enrolled Macs, and a list of existing macOS configuration policies to map for parity. Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. WS1/ABM specifics deferred to vendor docs. #> [CmdletBinding()] param()

How to run: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and these read scopes (consented at sign-in): DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementManagedDevices.Read.All. EXPECTED RESULTS APNs certificate presence/expiry, ADE/enrollment token(s) with synced macOS counts, count and version spread of enrolled Macs, and a list of existing macOS configuration policies to map for parity. Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. WS1/ABM specifics deferred to vendor docs. #> [CmdletBinding()] param()

WS1MAC-Remediation.ps1

Purpose: When WS1MAC-Validation / WS1MAC-Report shows a migrated Mac that enrolled but whose parity objects did not fully apply (or drifted), this issues a device sync action so the Intune settings-catalog policies, custom profiles, and shell scripts re-evaluate and re-apply, and confirms the Mac is still in its assigned migration ring group. It does NOT hand-edit settings on the Mac — Intune stays the single source of truth; on-device enforcement remains in the Apple .mobileconfig/shell controls. It does NOT reassign devices in Apple Business Manager or touch Workspace ONE; those are done in their consoles (deferred to vendor docs). HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\WS1MAC-Remediation.ps1 -ParityNameLike "WS1MAC Parity" Apply: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Remediation.ps1 -ParityNameLike "WS1MAC Parity" -WhatIf:$false Requires the Microsoft Graph PowerShell SDK and DeviceManagementConfiguration.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (for the sync action). EXPECTED RESULTS With -WhatIf (default): lists the drifted/stuck Macs and the sync actions it WOULD issue. With -WhatIf:$false: issues a sync to each so the parity objects re-apply. Re-run WS1MAC-Validation.ps1 after a policy cycle. .NOTES Migrations — Workspace ONE to Intune macOS. Defaults to -WhatIf (no change unless -WhatIf:$false). ABM/WS1 actions deferred to vendor consoles. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([string]$ParityNameLike =

'WS1MAC Parity')

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\WS1MAC-Remediation.ps1 -ParityNameLike "WS1MAC Parity" Apply: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Remediation.ps1 -ParityNameLike "WS1MAC Parity" -WhatIf:$false Requires the Microsoft Graph PowerShell SDK and DeviceManagementConfiguration.Read.All + DeviceManagementManagedDevices.PrivilegedOperations.All (for the sync action). EXPECTED RESULTS With -WhatIf (default): lists the drifted/stuck Macs and the sync actions it WOULD issue. With -WhatIf:$false: issues a sync to each so the parity objects re-apply. Re-run WS1MAC-Validation.ps1 after a policy cycle. .NOTES Migrations — Workspace ONE to Intune macOS. Defaults to -WhatIf (no change unless -WhatIf:$false). ABM/WS1 actions deferred to vendor consoles. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='High')] param([string]$ParityNameLike = 'WS1MAC Parity')

WS1MAC-Report.ps1

Purpose: Reports, across the Mac fleet or a migration ring, how each device evaluates against the named WS1MAC parity objects in Intune — by reading device statuses for the settings-catalog parity policies and custom configuration profiles, plus the run states of the parity shell scripts. Summarizes succeeded vs drifted (error/conflict/failed/notApplicable) per object, so you can confirm a ring reached parity before expanding. Read-only against Microsoft Graph (Intune). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WS1MAC-Report.ps1 -ParityNameLike "WS1MAC Parity" -CsvPath C:\Temp\ws1mac-parity.csv Requires the Microsoft Graph PowerShell SDK and DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A per-object breakdown of device states (succeeded/error/conflict/failed/etc.) and an optional per-device CSV showing which Macs have drifted from parity. Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. #> [CmdletBinding()] param([string]$ParityNameLike = 'WS1MAC Parity', [string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Report.ps1 -ParityNameLike "WS1MAC Parity" -CsvPath C:\Temp\ws1mac-parity.csv Requires the Microsoft Graph PowerShell SDK and DeviceManagementConfiguration.Read.All. EXPECTED RESULTS A per-object breakdown of device states (succeeded/error/conflict/failed/etc.) and an optional per-device CSV showing which Macs have drifted from parity. Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. #> [CmdletBinding()] param([string]$ParityNameLike = 'WS1MAC Parity', [string]$CsvPath)

WS1MAC-Set-MigrationRingTargeting.ps1

Purpose: Creates (if missing) the two migration ring security groups used throughout this solution — 'WS1MAC Migration - Pilot Ring' and 'WS1MAC Migration - Production Ring' — and adds the supplied Macs to the chosen ring. This is how you target which devices move in which wave. Group names are reused verbatim by the parity-object assignments in the Implementation Guide. It changes Entra group membership, so it defaults to -WhatIf (preview). It does NOT reassign devices in Apple Business Manager or unenroll from Workspace ONE — those are vendor-console actions (deferred to vendor docs). HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\WS1MAC-Set-MigrationRingTargeting.ps1 -Ring Pilot -DeviceNames "MAC-001","MAC-002" Apply: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Set-MigrationRingTargeting.ps1 -Ring Pilot -DeviceNames "MAC-001","MAC-002" -WhatIf:$false Requires the Microsoft Graph PowerShell SDK and

these scopes: Group.ReadWrite.All, DeviceManagementManagedDevices.Read.All, Device.Read.All. EXPECTED RESULTS With -WhatIf (default): reports which groups would be created and which Macs would be added. With -WhatIf:$false: creates the ring groups if missing and adds the named Macs to the chosen ring. Re-run WS1MAC-Validation.ps1 after assigning parity objects to confirm targeting. .NOTES Migrations — Workspace ONE to Intune macOS. Defaults to -WhatIf. ABM reassignment / WS1 release are vendor-console actions. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([ValidateSet('Pilot','Production')] [string]$Ring = 'Pilot', [string[]]$DeviceNames = @())

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\WS1MAC-Set-MigrationRingTargeting.ps1 -Ring Pilot -DeviceNames "MAC-001","MAC-002" Apply: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Set-MigrationRingTargeting.ps1 -Ring Pilot -DeviceNames "MAC-001","MAC-002" -WhatIf:$false Requires the Microsoft Graph PowerShell SDK and these scopes: Group.ReadWrite.All, DeviceManagementManagedDevices.Read.All, Device.Read.All. EXPECTED RESULTS With -WhatIf (default): reports which groups would be created and which Macs would be added. With -WhatIf:$false: creates the ring groups if missing and adds the named Macs to the chosen ring. Re-run WS1MAC-Validation.ps1 after assigning parity objects to confirm targeting. .NOTES Migrations — Workspace ONE to Intune macOS. Defaults to -WhatIf. ABM reassignment / WS1 release are vendor-console actions. #> [CmdletBinding(SupportsShouldProcess=$true, ConfirmImpact='Medium')] param([ValidateSet('Pilot','Production')] [string]$Ring = 'Pilot', [string[]]$DeviceNames = @())

WS1MAC-Validation.ps1

Purpose: For a given migrated Mac (or all Macs in a migration ring group), confirms it is enrolled and managed by Intune and that the named parity objects applied: the settings-catalog parity policy, custom .mobileconfig parity profiles, and shell-script parity. Reports each object's state for the device so you can confirm parity before expanding. Read-only against Microsoft Graph (Intune). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\WS1MAC-Validation.ps1 -DeviceName "MAC-001" powershell -ExecutionPolicy Bypass -File .\WS1MAC-Validation.ps1 -ParityNameLike "WS1MAC Parity" Requires the Microsoft Graph PowerShell SDK and these read scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. EXPECTED RESULTS For each matched Mac: enrolled/managed-by status, and per parity object (settings catalog, custom profile, shell script) the device's applied state (succeeded / error / not applied). Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. #> [CmdletBinding()] param([string]$DeviceName, [string]$ParityNameLike = 'WS1MAC Parity')

How to run: powershell -ExecutionPolicy Bypass -File .\WS1MAC-Validation.ps1 -DeviceName "MAC-001" powershell -ExecutionPolicy Bypass -File .\WS1MAC-Validation.ps1 -ParityNameLike "WS1MAC Parity" Requires the Microsoft Graph PowerShell SDK and these read scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. EXPECTED RESULTS For each matched Mac: enrolled/managed-by status, and per parity object (settings catalog, custom profile, shell script) the device's applied state (succeeded / error / not applied). Nothing is changed. .NOTES Migrations — Workspace ONE to Intune macOS. Read-only against Graph. #> [CmdletBinding()] param([string]$DeviceName, [string]$ParityNameLike = 'WS1MAC Parity')